Tuesday Reading Assessment: Chapters 6-1 through 6-6

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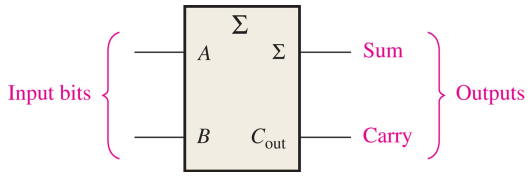


TABLE 6–1

Half-adder truth table.

A	*B*	C_{out}	Σ
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Σ = sum
C_{out} = output carry
A and *B* = input variables (operands)

Figure 1: (Top) Symbol for a *half-adder*, that adds two bits and produces the sum and carry bits. (Bottom) The truth table for the sum and carry bits.

1 Functions of Combinatorial Logic: Half-Adders

1. Examine the Σ (sum) output in the truth table in Fig. 1 (right). What gate accurately describes Σ in terms of the inputs A and B?

2. Examine the C_{out} (carry) output in the truth table in Fig. 1 (right). What gate accurately describes C_{out} in terms of the inputs A and B?

3. Create a logic circuit schematic in the space below that functions like the component in Fig. 1 (left), producing the truth table in Fig. 1 (right).

2 Functions of Combinatorial Logic: Full-Adders

1. Based on Sec. 1, we will now design a *full-adder* circuit, that includes the possibility of a carry-in bit. A carry-in occurs when a previous addition generates a carry-out bit, and that bit need to be added to the current calculation.

   - Write down your circuit for the half-adder below.

   - Now include gates and connections so that a new input, C_{in}, is added to the sum output.

   - Now include gates and connections so that the original output, C_{out}, accounts for C_{in}.